

The Impact of Hospital-Sponsored Health Plans on Readmission Rates

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PRELIMINARY

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Abstract

The healthcare sector has become increasingly vertically integrated, with hospital systems integrating healthcare financing by launching their own health plans. This paper investigates the causal effect of hospital health-plan vertical integration on hospital readmission using data on Medicare Advantage beneficiaries. We distinguish three causal objects: (i) the effect on patient outcomes of being enrolled in a provider-sponsored MA plan (PSHP), (ii) the effect of receiving care at a vertically integrated (VI) hospital, and (iii) the interaction of the two. For the plan channel, we adopt an instrumental variables approach that combines (a) the quasi-random differences in PSHPs availability within zip codes that span more than one county. For the hospital channel, we exploit an ambulance-based instrument for hospital assignment. We find that PSHP have lower readmission rate by 1.7% than standard plans but the difference fades as the hospital markets become more concentrated.

Keywords: Health Economics, Medicare Advantage, Vertical Integration, Provider-Sponsored Health Plan, Hospital Readmission.

1 Introduction

Health care market consolidation in the United States has evolved over time. Historically, consolidation consisted primarily of hospitals mergers and insurers mergers, while more recently hospital and insurers are becoming increasingly vertically integrated. This latter form of integration is not only the result of mergers and acquisitions, but comes in the form of large health systems that integrate the insurance business with healthcare delivery. In recent years, there has been a rise in hospitals and health systems offering their own health plans. As of 2019, approximately 37 percent of Medicare beneficiaries are enrolled in a MA plan and 20 percent of MA plans are sponsored by a health system. See 1-4.

The literature on hospital-insurer vertical integration is primarily focused the trade-off between efficiency and market foreclosure. Instead, little is know about the effect of hospital sponsored health plans on healthcare utilization and health outcomes. Proponents argue that the vertical integration of providers and insurers can lead to administrative and clinical integration, possibly achieving high efficiency and quality (Orszag). However, opponents suspect that profit motives are the driving force behind vertical integration and believe that the vertical integration of providers and insurers may increase health care utilization and costs without necessarily improving quality (Geruso).

Provider-sponsored health plans (PSHPs)—insurance plans sponsored by health systems that also deliver care—have become an increasingly important form of vertical integration in Medicare Advantage (MA). Proponents argue that tighter integration between insurer and provider can improve care coordination and reduce waste, while critics emphasize the potential for market power and limited competitive discipline. Despite growing policy interest, there remains limited evidence on whether hospital-sponsored plans translate into meaningfully different *patient outcomes* at the hospital level, and whether any differences depend on the competitive environment in which hospitals operate.

This paper studies one specific and policy-relevant dimension of quality: 30-day unplanned hospital readmissions following emergency admissions. Readmissions are costly for MA plans and Medicare, reflect both hospital care and discharge management, and are a central target of quality measurement and payment reforms. Using administrative data on emergency episodes, we document a robust descriptive pattern: patients treated at plan-sponsored hospitals have lower risk-adjusted readmission rates than patients treated at other hospitals, but the magnitude of this difference is smaller in more concentrated local hospital markets. Section ?? summarizes this descriptive evidence, and Section ?? presents an illustrative model that rationalizes these patterns through the interaction of internalized readmission costs and market concentration.

A key challenge is that descriptive comparisons need not have a causal interpretation. Even with rich clinical risk adjustment and beneficiary controls, unobserved patient severity, referral patterns, and plan- and provider-level selection may jointly influence both the destination hospital (plan-sponsored versus non-sponsored) and the probability of readmission. To address this identification problem, we use two quasi-experimental instruments commonly employed in the emergency care and hospital choice literature: (i) an ambulance-based instrument that exploits quasi-random assignment to ambulance providers with different propensities to transport patients to plan-sponsored hospitals; and (ii) a differential-distance instrument that captures plausibly exogenous variation in proximity to plan-sponsored versus non-sponsored hospitals. These instruments shift admission to plan-sponsored hospitals while holding fixed observed patient risk and local market conditions, allowing us to estimate causal effects for the subset of emergency patients whose destination hospital is influenced by emergency transport practices and/or proximity.

Our empirical framework is built around two estimands. First, we estimate the average causal effect of admission to a plan-sponsored hospital on readmissions for emergency episodes (a local average treatment effect for compliers). Second, we characterize how this causal effect varies with local hospital market concentration, motivated by the descriptive evidence and the model’s prediction that the incremental benefit of vertical integration attenuates in highly concentrated markets. We implement this heterogeneity analysis using predetermined concentration measures and market fixed effects, and report robustness using concentration bins.

This paper contributes to the literature on vertical integration and health plan ownership in three ways. First, it provides causal evidence on an outcome measure closely tied to quality and spending in MA: readmissions. Second, it documents and quantifies systematic heterogeneity in the causal effect of plan-sponsored hospitals across market structures. Third, by leveraging two complementary instruments within a unified framework and reporting transparent first-stage and balance diagnostics, it clarifies what can be learned about hospital behavior under integration from quasi-experimental variation in emergency hospital assignment.

Roadmap. Section ?? describes PSHPs and the institutional setting. Section ?? presents descriptive evidence on readmissions by hospital sponsorship status and market concentration. Section ?? introduces an illustrative model. Section 4 describes the empirical framework and identification strategy. Section ?? reports causal estimates and heterogeneity by concentration. Section ?? concludes.

2 Institutional Setting

2.1 Medicare Advantage and provider-sponsored plans

Medicare Advantage (MA) is the alternative to traditional fee-for-service Medicare and covers its beneficiaries through contracts with private plans. Integration between the provider and the health plan is becoming a prominent feature of the Medicare Advantage (MA) program. As MA plans are paid on a capitated basis and receive additional bonus payments based on quality performance, there may be an increased incentive to provide integrated care across care settings in order to maintain healthcare cost below the capitated amount and improve their enrollees' health to achieve the quality-based bonus.

Provider-sponsored health plans (PSHPs) are vertically integrated entities where a provider organization owns and operates its own insurance plan to provide healthcare services. This integration combines the payer-side (insurance) and provider-side (hospitals, doctors) functions within a single organization. This organization often takes the form of a health system, but there are instances of health plans that are sponsored by provider organizations that do not include hospitals.

2.2 Outcome: 30-day unplanned hospital readmission

This paper focuses on hospital readmissions as a clinically meaningful and policy-relevant measure of care coordination and discharge quality. Readmissions are financially consequential for Medicare and hospitals, and they are explicitly targeted by Medicare's Hospital Readmissions Reduction Program (HRRP). Recent evidence indicates that a substantial share of acute admissions are followed by a rehospitalization within 30 days, with aggregate spending associated with readmissions on the order of tens of billions of dollars annually in Medicare.¹

Definition and construction. Our main outcome, RR_{it} , is an indicator equal to one if beneficiary i experiences an *unplanned* acute inpatient readmission within 30 days of discharge from the *index* emergency admission at time t , and zero otherwise. We treat multiple rehospitalizations within the 30-day window as a single event, consistent with HRRP-style targeted measures, which are most naturally interpreted as readmission probabilities rather than counts.² Following CMS-style measurement conventions, we exclude same-day transfers and implement a planned-readmission algorithm to remove rehospitalizations that are

¹See, e.g., ? for summary statistics and policy discussion.

²? notes that CMS does not differentiate between one and multiple readmissions within the 30-day period when constructing targeted readmission metrics.

predictably scheduled (e.g., selected elective procedures) rather than reflecting failures of post-discharge management.³ Because our data are Medicare Advantage encounters, we can track readmissions to *any* acute-care hospital, not only the index hospital.

All specifications adjust for rich clinical risk at the patient level. In addition to demographics and a baseline risk score, we include Hierarchical Condition Category (HCC) comorbidity indicators and principal-diagnosis fixed effects for the index admission. This aligns with the spirit of CMS risk adjustment, which uses case-level models with extensive risk factors when computing risk-adjusted readmission measures.⁴

Why readmissions are informative for this setting. The policy motivation for using readmissions is that a non-trivial share of rehospitalizations may be avoidable through improved care transitions and coordination. In our context, this maps naturally to organizational features that affect discharge planning, follow-up management, and the integration of inpatient care with post-acute and outpatient settings.

Financial incentives and the HRRP. HRRP, implemented beginning in 2013, links hospital revenue to readmission performance. CMS evaluates hospital readmission performance over a multi-year window and applies payment reductions to hospitals with “excess” readmissions relative to national benchmarks.⁵ The program initially targeted acute myocardial infarction, heart failure, and pneumonia, and later expanded to additional conditions and procedures.⁶ Penalties were phased in from a 1% cap in 2013 to 2% in 2014 and 3% from 2015 onward.⁷ Thus, readmissions affect hospital financial performance both directly (through costly utilization) and indirectly through explicit Medicare payment reductions tied to risk-adjusted readmission rates.⁸

3 Data and Motivating Evidence

Medicare Advantage Claims Data We use administrative data from the Centers for Medicare & Medicaid Services (CMS) for the years 2019 and 2022. The main data source for plan enrollment is the Master Beneficiary Summary File (MBSF), which contains individual-

³? describe the 30-day all-cause framework and common exclusions used in HRRP-style definitions.

⁴? describes CMS’s use of hierarchical logistic regression with a large set of patient risk factors for condition-specific readmission measures.

⁵? provides a detailed description of the HRRP penalty structure, including the use of risk-adjusted readmission rates over a three-year window and benchmarking against the national mean.

⁶? summarizes the targeted conditions/procedures and the program timeline.

⁷?

⁸? discusses how excess readmission rates translate into payment reductions under HRRP.

level data on a 5% random sample of Medicare beneficiaries. The MBSF records enrollment information for each beneficiary, including whether they are enrolled in Traditional Medicare or a Medicare Advantage (MA) plan. The MBSF also includes detailed beneficiary characteristics such as age, gender, race, income measures, and county of residence. In 2019, total Medicare enrollment was approximately 61 million, so the 5% sample includes about 3 million individuals. Roughly 37% of beneficiaries were enrolled in MA plans. I exclude ESRD patients from the analysis and also special needs plans, pace plans, employer-sponsored plans or PFFS plans which have a different reimbursement scheme from CMS which is different from the standard plans. The main data source for the patients' health claims is the Inpatient Encounter Data, from which we can obtain the ambulance company and the hospital identifiers where patients were treated, the ICD-10 diagnosis codes, the procedure codes, the length of hospital stay, and the discharge location. In addition, I use the Inpatient, Outpatient, Skilled Nursing Facility (SNF), Home Health Agency (HHA), Durable Medical Equipment (DME), and Carrier Encounter files to calculate the average cost of care for Medicare beneficiaries.

Provider-Sponsored Health Plans To identify integrated MA plans sponsored by health systems, we used the AHRQ Compendium of Health Systems which contains the list of all health plans offered by hospitals and health systems. We then linked all of these hospitals to their CMS provider identification numbers (CCN) to identify them in AHA database and add information on hospital characteristics. To identify health plans sponsored by other provider organizations, we searched through the full list of MA contracts to flag those whose sponsor is a provider organization.

3.1 Descriptive Evidence

A central premise of the paper is that hospital-insurer vertical integration can reduce readmissions by strengthening incentives and capabilities for care coordination. In the illustrative model, greater internalization of the plan-side margin leads hospitals to choose higher care-coordination effort and therefore lower readmission risk. Before turning to the instrumental variables design, this section presents suggestive (non-causal) evidence consistent with that mechanism and highlights an additional empirical pattern: the association between vertical integration and readmissions is strongest in competitive markets and attenuates as local hospital markets become more concentrated.

We study how the within-hospital association between vertical integration and risk-adjusted readmissions varies with hospital market concentration. Let m index the hospital's

local market (HSA in the main specification). We estimate:

$$\text{RR}_{het} = \beta_1 \text{VIHosp}_{ht} + \beta_2 \ln(\text{HHI})_{mt} + \beta_3 \text{VIHosp}_{ht} \times \ln(\text{HHI})_{mt} + X'_{het}\gamma + \delta_e + \tau_t + \varepsilon_{het}, \quad (1)$$

where RR_{het} is the risk-adjusted 30-day readmission rate for hospital h , episode type e , and year t ; VIHosp_{ht} indicates whether the admitting hospital is vertically integrated with a plan sponsor. We consider two market definitions, Hospital Service Area (HSA) and County Area, and compute the corresponding log Herfindahl–Hirschman Index ($\ln(\text{HHI})_{mt}$) using acute-care bed shares. In the estimation $\ln(\text{HHI})_{mt}$ is mean-centered to ease interpretation. We include a set of risk adjusters and control for a rich set of hospital and market characteristics, along with episode fixed effects δ_e and year fixed effects τ_t . Regressions are weighted by $\ln(\text{discharges})$ and standard errors are clustered at the hospital (CCN) level.

Table 1 summarizes the key coefficients. Two facts stand out. First, at the average level of concentration (i.e., when $\ln(\text{HHI})_{mt} = 0$), vertically integrated hospitals have lower readmission rates than otherwise similar non-integrated hospitals ($\beta_1 < 0$). Second, the interaction term is positive and statistically significant ($\beta_3 > 0$), implying that the VI–non-VI readmission gap becomes smaller as hospital markets become more concentrated.

At the sample mean level of market concentration, vertically integrated (VI) hospitals are associated with *lower* risk-adjusted readmission rates. Because $\ln(\text{HHI})$ is mean-centered, the coefficient on **VI hospital** captures the VI–non-VI difference evaluated at the **average** HHI in the estimation sample: in column (1) (HSA markets) the estimate -0.297 implies that VI hospitals have readmission rates that are 0.297 percentage points lower than non-VI hospitals at mean HHI; the analogous estimate in column (2) (county markets) is -0.288 percentage points. The positive and statistically significant interaction term, **VI hospital** $\times \ln(\text{HHI})$, implies that this VI advantage *shrinks* as market concentration increases. Using the point estimates, a 10% increase in HHI attenuates the VI–non-VI readmission gap by approximately 0.034 pp in HSA market and

Using the point estimates, a 10% increase in HHI relative to the mean reduces the magnitude of the VI effect by about $0.355 \times \ln(1.10) \approx 0.034$ in column (1) and $0.297 \times \ln(1.10) \approx 0.028$ in column (2). Interpreting these in percentage terms, this corresponds to an attenuation of roughly $100 \times (\exp(0.034) - 1) \approx 3.5\%$ (HSA) and $100 \times (\exp(0.028) - 1) \approx 2.9\%$ (county) in the VI–non-VI readmission gap.

Therefore, the main pattern we observe is that VI-hospitals are associated with lower level of risk adjusted readmission rate compared to standard hospitals, but this association fades away as the level of market concentration increases, and vanished in highly concentrated markets. However, we are not interpreting these results are causal since hospital

Table 1: Readmission rates, vertical integration, and market concentration

	(1) Market = HSA	(2) Market = County
VI hospital	-0.297*** (0.105)	-0.288*** (0.104)
ln(HHI)	-0.135 (0.190)	-0.111 (0.178)
VI hospital $\times$ ln(HHI)	0.355*** (0.135)	0.297*** (0.114)
HS hospital	-0.365*** (0.099)	-0.378*** (0.099)
HS hospital $\times$ ln(HHI)	0.034 (0.140)	0.155 (0.105)
ln(Expected RR)	14.879*** (0.601)	15.045*** (0.599)
ln(Risk score)	2.636*** (0.460)	2.790*** (0.463)
Episode FE	Yes	Yes
Year FE	Yes	Yes
Weights	ln(discharges)	ln(discharges)
Clusters (CCN)	1,941	1,941
Observations	85,610	75,679

Notes: Unit of observation is hospital $\times$ episode type $\times$ year. The dependent variable is the risk-adjusted 30-day readmission rate. ln(HHI) is the log Herfindahl–Hirschman Index constructed from acute-care bed shares and mean-centered in the estimation sample. Markets are defined as HSAs in column (1) and counties in column (2). Both specifications include episode and year fixed effects and analytic weights equal to ln(discharges). A rich set of hospital and market-level controls are included but not reported. Standard errors (in parentheses) are clustered at the hospital (CCN) level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

integration status is endogenous, and unobserved differences across integrated and non-integrated hospitals (or across markets) could drive the observed gradient. This motivates the instrumental-variables design in the next section, which isolates plausibly exogenous variation in admission to a VI hospital and enrollment in PSHP.

Figure 1 provides a transparent visualization of this relationship. The figure residualizes readmission rates on the full set of controls and year/episode fixed effects, then plots binned means against HSA-level HHI separately for VI and non-VI hospitals, along with within-group linear fits. The VI series starts below the non-VI series in competitive HSAs, but rises more steeply with HHI, so the vertical gap narrows in highly concentrated HSAs.

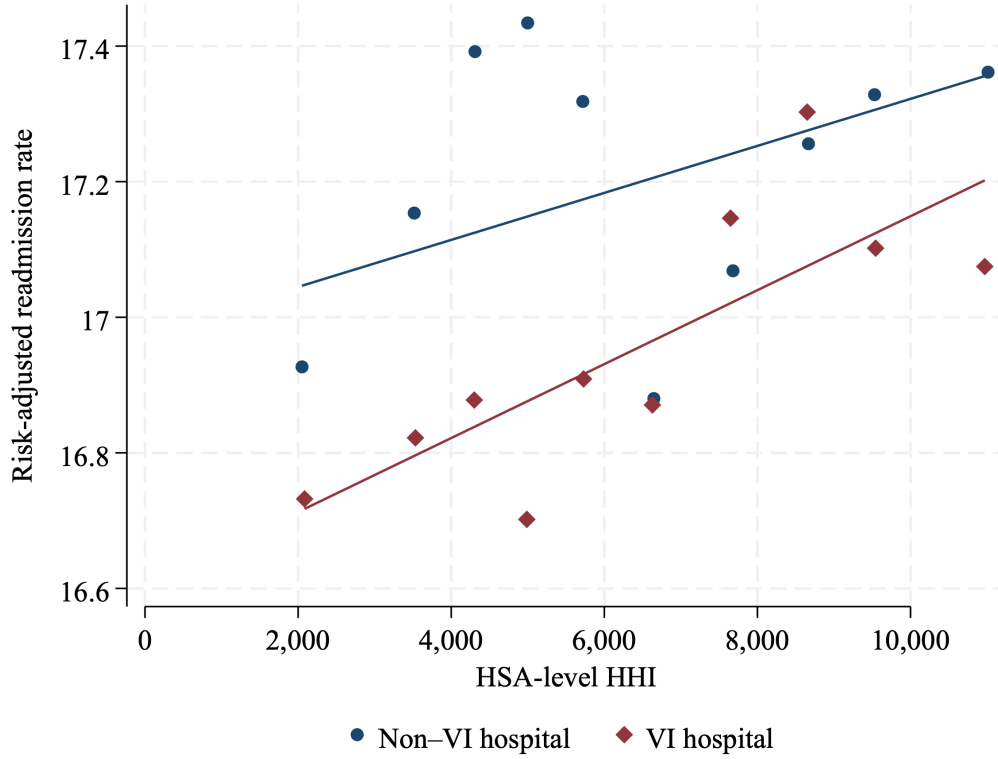


Figure 1: Readmission Rate vs. HHI by VI Hospital Status

Notes: Each point shows the mean readmission rate within each decile of the HSA-level HHI distribution, separately for VI and non-VI hospitals. The plotted outcome is residualized on a rich set of hospital and market controls, year fixed effects, and episode indicators, using $\ln(\text{discharges})$ as analytic weights. The HSA-level HHI is constructed from acute-care bed shares. Lines are within-group linear fits.

Why OLS is not causal: patient selection into hospitals A naïve comparison of readmissions between plan-sponsored and non-sponsored hospitals can be confounded by selection. In emergency care settings, hospital assignment is influenced by ambulance routing protocols, proximity, and network considerations, but it may also respond to unobserved patient severity and clinical complexity not fully captured by claims-based risk adjustment. For example, patients with more complex discharge needs may be preferentially transported to particular hospitals; similarly, unobserved social risk and post-acute care access may affect both hospital choice and readmissions. These unobservables generate correlation between D_i and the structural error term in outcome equations, biasing ordinary least squares (OLS) estimates. The empirical strategy therefore uses instrumental variables to generate quasi-random variation in admission to plan-sponsored hospitals.

3.2 Illustrative Model

To rationalize the reduced-form evidence, this section presents a simple illustrative model of hospital effort to prevent readmissions. The model delivers two qualitative implications consistent with the empirics: (i) vertically integrated (plan-sponsored) hospitals exert more readmission-prevention effort than otherwise similar non-integrated hospitals; and (ii) the incremental effort advantage from vertical integration attenuates in more concentrated markets.

Consider a single index admission for a Medicare Advantage (MA) beneficiary treated at a hospital. After discharge, the beneficiary is readmitted with probability $p(e)$, which depends on the hospital's effort $e \geq 0$ (e.g., discharge planning, follow-up, care coordination). For transparency, assume a linear readmission technology

$$p(e) = \bar{p} - e, \quad e \in [0, \bar{p}], \quad \bar{p} \in (0, 1), \quad (2)$$

so $p'(e) = -1 < 0$. Effort is an *upfront* investment (organizational capabilities, staffing, protocols) with quadratic cost

$$C(e) = \frac{c}{2}e^2, \quad c > 0. \quad (3)$$

Baseline hospital profits under negotiated MA fees. Let $s_0 \in [0, 1]$ denote the hospital's baseline market share in the local hospital market (e.g., measured using beds or discharges). For patients enrolled in *other* MA plans (i.e., not the hospital's sponsor plan), the hospital is paid negotiated fees that may depend on local market conditions. Let $F^{NR}(s_0)$ and $F^R(s_0)$ denote the negotiated payments in the no-readmission and readmission states, respectively, and let w^{NR} and w^R be the corresponding treatment costs, with $w^R > w^{NR}$. Define per-episode profits

$$\pi^{NR}(s_0) \equiv F^{NR}(s_0) - w^{NR}, \quad \pi^R(s_0) \equiv F^R(s_0) - w^R, \quad (4)$$

and assume readmissions are privately undesirable for the hospital:

$$\Delta\pi(s_0) \equiv \pi^{NR}(s_0) - \pi^R(s_0) > 0. \quad (5)$$

A standard (non-integrated) hospital chooses e to maximize expected profits from its baseline volume, proportional to s_0 :

$$\Pi^S(e; s_0) = s_0 \left[(1 - p(e))\pi^{NR}(s_0) + p(e)\pi^R(s_0) \right] - \frac{c}{2}e^2. \quad (6)$$

Using $p'(e) = -1$, the first-order condition (FOC) is

$$\frac{\partial \Pi^S}{\partial e} = s_0 \left(\pi^{NR}(s_0) - \pi^R(s_0) \right) - ce = s_0 \Delta \pi(s_0) - ce = 0, \quad (7)$$

so the optimal effort for a standard hospital is

$$e^S(s_0) = \frac{s_0 \Delta \pi(s_0)}{c}. \quad (8)$$

Plan-sponsored (vertically integrated) hospital. Now consider a hospital that sponsors an MA plan. Let $\mu \in [0, 1]$ denote the share of MA beneficiaries in the local market who are enrolled in the sponsor plan. The sponsor plan receives a capitated payment b from Medicare for each beneficiary enrolled in its plan, *regardless of where enrollees are treated*. Thus, the hospital revenue of from its plan enrollees is μb .

The key incentive difference under vertical integration is that the integrated entity internalizes the medical cost consequences of readmissions for the subset of sponsor-plan enrollees whose care is delivered in-system and therefore affected by the hospital's effort. Let $(1 - s_0)h(\mu)$ denote the incremental mass of such "in-system" episodes induced by plan sponsorship, where $h(\cdot)$ is increasing and $h(0) = 0$. The factor $(1 - s_0)$ captures saturation: when the hospital already has high baseline market share (high concentration / strong provider market power), sponsoring a plan adds less incremental exposure to the sponsor system's care-management technology.

Define the expected (resource) cost of an episode as

$$\tilde{w}(e) \equiv (1 - p(e))w^{NR} + p(e)w^R. \quad (9)$$

Using (2), we have

$$\tilde{w}'(e) = w^{NR} - w^R = -(w^R - w^{NR}) \equiv -\Delta w, \quad \Delta w > 0. \quad (10)$$

The plan-sponsored hospital's objective is

$$\Pi^{VI}(e; s_0, \mu) = s_0 \left[(1 - p(e))\pi^{NR}(s_0) + p(e)\pi^R(s_0) \right] + \mu b - (1 - s_0)h(\mu)\tilde{w}(e) - \frac{c}{2}e^2. \quad (11)$$

Differentiating and using (10) yields the FOC

$$\begin{aligned} \frac{\partial \Pi^{VI}}{\partial e} &= s_0 \Delta \pi(s_0) - (1 - s_0)h(\mu)\tilde{w}'(e) - ce \\ &= s_0 \Delta \pi(s_0) + (1 - s_0)h(\mu)\Delta w - ce = 0. \end{aligned} \quad (12)$$

Therefore, the optimal effort under vertical integration is

$$e^{VI}(s_0, \mu) = \frac{s_0 \Delta \pi(s_0) + (1 - s_0) h(\mu) \Delta w}{c}. \quad (13)$$

Model Predictions. Combining (8)–(13), the effort gap is

$$\Delta e(s_0, \mu) \equiv e^{VI} - e^S = \frac{(1 - s_0) h(\mu) \Delta w}{c}. \quad (14)$$

Since $h(\mu) \geq 0$ and $\Delta w > 0$, (14) implies:

1. **Vertical integration reduces readmissions.** For $\mu > 0$, $\Delta e(s_0, \mu) > 0$, so $e^{VI} > e^S$ and therefore $p(e^{VI}) < p(e^S)$ under (2). In words, plan sponsorship induces higher readmission-prevention effort because the integrated entity internalizes the medical-cost savings from avoiding readmissions for an additional mass of sponsor-plan episodes.
2. **The VI advantage attenuates with concentration.** Holding μ fixed, (14) implies $\Delta e(s_0, \mu) \rightarrow 0$ as $s_0 \rightarrow 1$. Intuitively, in highly concentrated markets the sponsor hospital is already a must-have provider (high baseline share), so plan sponsorship generates little additional in-system exposure for which the integrated entity internalizes readmission costs; as a result, the incremental effort incentive from vertical integration vanishes.

These implications mirror the empirical pattern that VI hospitals are associated with lower risk-adjusted readmission rates, but the magnitude of this association fades as local hospital markets become more concentrated.

4 Empirical Framework

The unit of observation is an emergency care episode indexed by i , defined as an emergency admission (or ED-to-inpatient transfer) for an MA beneficiary in year $t(i)$. The primary outcome is an indicator for 30-day unplanned readmission,

$$y_i \in \{0, 1\}.$$

Let $h(i)$ denote the admitting hospital. The endogenous treatment is an indicator for whether the admitting hospital is plan-sponsored:

$$D_i \equiv \mathbb{1}\{h(i) \text{ is a plan-sponsored (PSHP) hospital}\}.$$

We condition on a rich vector of patient and episode controls X_i (e.g., demographics, risk score / HCCs, comorbidity flags, diagnosis or condition fixed effects, and time controls), and on plan controls (see below).

Markets are indexed by m , constructed at the county (or other pre-specified) level. Let $m(i)$ denote the market in which hospital $h(i)$ operates. We measure market concentration using a predetermined Herfindahl–Hirschman Index (HHI) constructed from hospital capacity or utilization shares in a pre-period:

$$M_m \equiv \ln(\text{HHI}_m),$$

where higher M_m corresponds to more concentrated markets. Because M_m is predetermined and varies only at the market level, we include market fixed effects α_m in all baseline specifications; these fixed effects absorb the level effect of M_m and other time-invariant market characteristics.

We estimate a linear probability model with market fixed effects and allow the causal effect of admission to a plan-sponsored hospital to vary with market concentration. The structural equation is

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 (D_i \times \tilde{M}_{m(i)}) + X_i' \gamma + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i)} + \varepsilon_i, \quad (15)$$

where $\tilde{M}_m \equiv M_m - \bar{M}$ is the mean-centered concentration index, τ_t are year fixed effects, and $\phi_{\text{plan}(i)}$ are plan fixed effects.⁹ Mean centering implies that β_1 is the causal effect evaluated at the average concentration level in the sample, while β_2 captures the slope of heterogeneity with respect to concentration. Because M_m is absorbed by market fixed effects, we do not interpret concentration causally; rather, $\tilde{M}_m$ indexes how the VI-hospital effect varies systematically across market structures.

Fixed effects and their role in identification. The fixed effects in (15) serve two purposes. First, market fixed effects α_m absorb all time-invariant differences across local hospital markets that could simultaneously affect (i) the probability of admission to a plan-sponsored hospital and (ii) baseline readmission risk, such as persistent differences in provider quality, practice style, post-acute care supply, local socioeconomic conditions, and the general level of hospital market concentration.¹⁰ Put differently, with α_m in place, the remaining

⁹In alternative specifications, we replace plan fixed effects with contract-by-year fixed effects to absorb time-varying differences in benefit design and utilization management at the contract level. In that case, we retain year fixed effects to capture common time shocks not absorbed by the contract-by-year controls.

¹⁰When concentration is measured using a predetermined market-level HHI that is constant (or nearly constant) over the study period, α_m also absorbs the level effect of $\ln(\text{HHI}_m)$, ensuring that identification

identifying variation comes from differences in patients’ admitting hospitals *within the same market*, holding fixed all market-level confounders that do not vary over time.

Second, plan fixed effects $\phi_{\text{plan}(i)}$ absorb persistent differences across MA plans that could affect readmissions directly (e.g., utilization management intensity, care coordination programs, network design, and enrollee composition) and may also influence which hospitals enrollees are likely to use. Year fixed effects τ_t absorb common time shocks affecting readmissions and hospital practice (e.g., coding changes, broad shifts in clinical protocols, or policy changes affecting all markets). In specifications with contract-by-year fixed effects, we allow these plan-level differences to vary flexibly over time at the contract level, further tightening identification by absorbing contract-specific year-to-year changes in benefits or care management that could otherwise contaminate comparisons across hospital types.

Together, these fixed effects substantially reduce scope for confounding from time-invariant market characteristics and persistent plan features, so that the remaining endogeneity concern is primarily *patient selection into hospitals within market and within plan*. This motivates the instrumental-variables strategy described in Section ??.

Throughout, standard errors are clustered at a level appropriate for the assignment mechanism (e.g., at the ambulance provider level for the ambulance IV sample, and at the residential ZIP or market level for the distance IV), with robustness to alternative clustering choices.

4.1 Endogeneity and instruments

The hospital level VI integration status could be endogenous for several reasons: (i) beneficiaries may select into VI hospitals based on unobserved preferences or health risk, and (ii) even in emergencies, unobserved severity and local provider attributes may correlate with being admitted to an integrated hospital. We therefore use an IV strategy with excluded instruments that generate quasi-experimental variations in the VI-hospital admission.

Ambulance-preference shifter for VI-hospital admission. The locus of treatment for emergency hospitalizations is, to a large extent, determined by pre-hospital factors, including ambulance transport decisions and patient location. Moreover, there is wide evidence that large city areas are often served by multiple ambulance companies, whereas smaller rural with a single ambulance company, neighboring companies provide service when the principal ambulance units are busy under so-called “mutual aid” agreements. Thus, in both cases,

of β_2 relies on within-market variation in D_i interacted with cross-market differences in $\tilde{M}_m$, rather than on levels comparisons across markets.

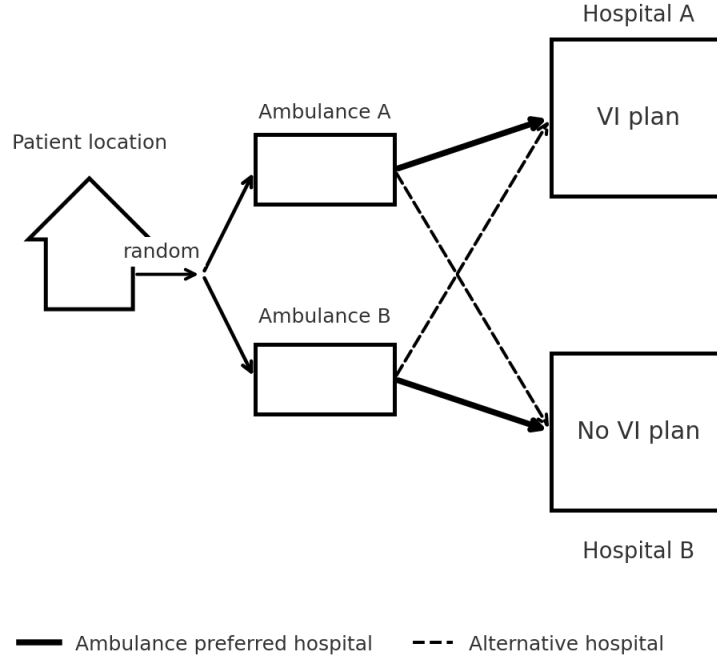


Figure 2: Hospital Assignment and Ambulance Preferences

the assignment of the ambulance company to the patient is effectively random. (Doyle 2015, Chiang, et al., 2006 Ragone 2012).

In addition to plausibly exogenous assignment, ambulance companies are expected to have preferences for particular hospitals. In survey work described in Doyle et al. (2015), we found that paramedics have developed relationships with local emergency departments. For example, Skura (2001) studied ambulance assignment in the wake of a new system of competition between public and private ambulances in New York City. He found that patients living in the same ZIP code as public Health and Hospital Corporation hospitals were less than half as likely to be taken there when assigned a private, non-profit ambulance (29%) compared to when the dispatch system assigned them to an FDNY ambulance (64%). In most cases, the private ambulances were operated by non-profit hospitals and stationed near or even within those facilities, so they tended to take their patients to their affiliated hospitals. More broadly, with the exception of acute trauma care for which there are often defined local protocols for hospital assignment (Kahn et al. 2008) transport assignment for other emergencies is more likely to be driven by idiosyncratic preferences. Thus, in this context patient transport decisions are more likely to be made in ways that are not systematically tied to the underlying health of the patient.

Figure 2 illustrates the key source of quasi-random variation in hospital assignment that

underlies the empirical strategy. When a medical emergency occurs, the dispatch system typically assigns a patient to the nearest available ambulance, which is effectively random from the patient’s perspective. Conditional on the patient’s location, the ambulance dispatched may have systematically different “preferred” hospitals due to historical routing patterns, contractual relationships, or local familiarity.

As shown in Figure 2, two ambulances (A and B) serve the same geographic area but exhibit different propensities to transport patients to different hospitals. Ambulance A tends to deliver patients to Hospital A, which is vertically integrated (VI) with a Medicare Advantage plan, whereas Ambulance B tends to deliver patients to Hospital B, which is not integrated. Importantly, ambulances occasionally transport patients to the alternative hospital as well, generating natural variation in destination choice.

This setup creates a quasi-experimental shock in exposure to VI hospitals: patients who happen to be picked up by an ambulance with stronger ties to the VI hospital are more likely to be admitted there, even though the ambulance choice is conditionally random. This type of variation has been widely used in the hospital choice and emergency-care literature because it mimics an instrumental variable: ambulance identity predicts hospital assignment but is plausibly orthogonal to patient unobservables.

In other words, patients typically do not choose the ambulance, and dispatch/rotation protocols and ambulance company service areas generate quasi-random assignment to companies with heterogeneous hospital destination preferences. This yields plausibly exogenous variation in the admitting hospital conditional on residential location and origin, forming the basis for ambulance preference instruments (Doyle Jr et al., 2015).

Following the emergency-care literature, we use an ambulance-company “preference” instrument that captures the propensity of ambulance company a to deliver patients to integrated hospitals. For each ambulance company a , market, year, and emergency condition, we compute the leave-one-out share of transports ending at an integrated hospital among other patients served by that company:

$$\text{AmbShare}_{it} = \frac{1}{N_{a(i),t,c(i)} - 1} \sum_{\substack{j:a(j)=a(i) \\ j \neq i}} \mathbf{1}\{\text{VI}_{jt}^h = 1\}, \quad (16)$$

where the summation is computed within the same market-year-emergency cell (to preserve relevance and comparability across episodes). Conditional on controls and fixed effects, we assume ambulance assignment affects outcomes only through the admitting hospital (and thus exposure to VI^h), not through unobserved severity or differential pre-hospital treatment.

Differential distance between VI and non-VI hospitals. We additionally use a geographic shifter based on relative proximity to integrated versus non-integrated hospitals. Let $\text{dist}(r(i), h)$ denote distance from the beneficiary’s residence (or pickup location) $r(i)$ to hospital h . Define the nearest plan-sponsored hospital $h^{VI}(i)$ and the nearest non-sponsored hospital $h^{NVI}(i)$, and construct

$$Z_i^{DD} \equiv \ln \text{dist}(r(i), h^{VI}(i)) - \ln \text{dist}(r(i), h^{NVI}(i)).$$

More negative values of Z_i^{DD} indicate greater relative proximity to a plan-sponsored hospital. The identifying intuition is that, in acute care settings, *distance is a first-order determinant of hospital assignment*: conditional on emergency condition, location, and market characteristics, patients (and ambulance routing protocols) tend to be transported to closer hospitals, and marginal differences in relative proximity shift the likelihood that the admitting hospital is plan-sponsored. In our setting, Z_i^{DD} therefore generates quasi-experimental variation in D_i by shifting the hospital set that is practically feasible for emergency admission, holding fixed rich patient risk adjustment and fixed effects.

Our exclusion restriction is that, conditional on the control set and fixed effects in (15), differential distance affects readmissions only through admission to a plan-sponsored hospital (and, in specifications that include plan controls, through plan enrollment channels already absorbed by those controls), not through unobserved severity or neighborhood-level confounders. The emergency/ambulance sample restrictions described below are designed to strengthen this restriction by limiting scope for discretionary hospital selection and for systematic differences in unobservables across beneficiaries living closer to VI versus non-VI hospitals.

Estimation samples and use of instruments. We construct the analysis sample in several steps. First, we restrict attention to non-discretionary emergency conditions—such as acute myocardial infarction, stroke, and hip fracture—that represent high-acuity clinical presentations for which hospital choice is effectively non-deferrable and largely outside the patient’s control. These conditions are clinically significant and account for a substantial share of inpatient utilization among Medicare beneficiaries.

Second, for the primary analysis we restrict the estimation sample to emergency hospitalizations arriving via ambulance transport. This restriction is central for the ambulance-based instrument (which relies on quasi-random assignment to ambulance providers with different destination propensities) and also strengthens the credibility of the differential-distance instrument by limiting scope for discretionary hospital selection by patients and families.

Applying these restrictions yields a beneficiary-by-episode dataset. The unit of observation is an emergency episode for beneficiary i in calendar year t , defined as an inpatient admission originating in the emergency department (or an observation stay subsequently converted to inpatient) with no transfer-in from another acute-care facility. For each episode, we observe the beneficiary’s MA plan enrollment at time t , the identity and characteristics of the admitting hospital, and utilization outcomes during and after the inpatient stay, including 30-day unplanned readmissions.

Our baseline IV estimates are obtained on the ambulance-transported emergency sample using the ambulance preference shifter as the primary instrument and, where informative, the differential-distance shifter as a complementary source of variation. Using both instruments on the same high-acuity ambulance sample keeps the identifying assumptions aligned across IV strategies and facilitates direct comparison of estimates.

Because differential distance does not mechanically require ambulance transport, we also report a robustness specification that expands the sample to *all* non-discretionary emergency admissions (including self-transport) and uses Z_i^{DD} as the excluded instrument. This broader emergency sample increases external validity and statistical power, while preserving the key feature that admissions are plausibly non-deferrable. In contrast, we do not use elective or discretionary (non-emergency) admissions as a primary estimation sample for the distance instrument, because hospital choice in such settings is more likely to reflect unobserved patient preferences, provider referrals, and plan network design, which can undermine the exclusion restriction for Z_i^{DD} . We therefore treat the emergency-based distance-IV estimates as the appropriate complement to the ambulance-IV design.

4.2 2SLS with heterogeneous effects

The presence of heterogeneity implies two endogenous regressors in (15): the indicator for admission to a plan-sponsored hospital D_i and its interaction with market concentration $D_i \times \tilde{M}_{m(i)}$. We therefore instrument the endogenous regressor vector

$$W_i \equiv \left(D_i D_i \times \tilde{M}_{m(i)} \right)$$

using the excluded instruments

$$Z_i \equiv \left(Z_i^A, Z_i^{DD}, Z_i^A \times \tilde{M}_{m(i)}, Z_i^{DD} \times \tilde{M}_{m(i)} \right),$$

along with the full set of included controls and fixed effects in (15).

What variation identifies each parameter. The coefficient β_1 is identified by instrument-induced shifts in admission to plan-sponsored hospitals that are orthogonal to unobserved patient severity, after conditioning on X_i and fixed effects. In practice, this variation comes from (i) quasi-random assignment to ambulance providers with different propensities to transport patients to plan-sponsored hospitals (captured by Z_i^A), and/or (ii) plausibly exogenous relative proximity to plan-sponsored versus non-sponsored hospitals (captured by Z_i^{DD}). Since $\tilde{M}_{m(i)}$ is predetermined and varies only at the market level (and is mean-centered), interacting instruments with $\tilde{M}_{m(i)}$ yields additional excluded variation that shifts the concentration-weighted treatment $D_i \times \tilde{M}_{m(i)}$. Intuitively, the same quasi-experimental shocks that move D_i generate larger movements in $D_i \times \tilde{M}_{m(i)}$ in more concentrated markets (higher $\tilde{M}_m$) and smaller movements in less concentrated markets (lower $\tilde{M}_m$). This interaction variation identifies β_2 , the slope of the treatment effect with respect to market concentration.

LATE interpretation. Under the standard IV assumptions (relevance, exclusion, and monotonicity), 2SLS identifies local average treatment effects (LATEs) for *compliers*: emergency patients whose destination hospital (VI vs. non-VI) is shifted by the instrument. With mean-centered $\tilde{M}_m$, the structural equation (15) can be written as

$$y_i = \beta_0 + (\beta_1 + \beta_2 \tilde{M}_{m(i)}) D_i + \dots,$$

so the implied causal effect of admission to a plan-sponsored hospital in market m is $\tau(m) \equiv \beta_1 + \beta_2 \tilde{M}_m$. The parameter β_1 should therefore be interpreted as the average complier effect evaluated at the mean concentration level in the sample (i.e., $\tilde{M}_m = 0$), while β_2 captures how the complier effect varies systematically with market concentration. Equivalently, β_2 is the incremental change in the IV-identified treatment effect associated with a one-unit increase in $\tilde{M}_m$.

Instrument validation and robustness. First-stage strength, balance on predetermined characteristics, and robustness exercises—including nonparametric heterogeneity using concentration bins and alternative samples (e.g., restricting to non-PSHP enrollees)—are reported in Appendix ??.

5 Preliminary Results

References

Doyle Jr, J. J., Graves, J. A., Gruber, J., and Kleiner, S. A. (2015). Measuring returns to hospital care: Evidence from ambulance referral patterns. *Journal of Political Economy*, 123(1):170–214.

Appendix A

5.1 Instrument validity and robustness checks

This section reports two core diagnostics: instrument relevance (first stage strength) and balance on predetermined covariates. In addition, we implement a robustness check that estimates heterogeneity nonparametrically using concentration bins.

5.1.1 Instrument relevance

We report first-stage estimates for both endogenous regressors, D_i and $D_i \times \tilde{M}_{m(i)}$. For each instrument set (ambulance only, distance only, and the combined set), we present the first-stage coefficients and associated robust weak-instrument diagnostics, including the Kleibergen–Paap rk Wald F statistic. Because the model includes multiple endogenous regressors, we report first-stage strength for each equation as well as the joint strength of the excluded instruments for the endogenous regressor vector.

5.1.2 Balance on predetermined characteristics

To assess whether the instruments behave as-if random conditional on the included controls and fixed effects, we conduct balance tests on predetermined patient and episode characteristics. Specifically, we estimate auxiliary regressions of the form

$$X_{ik} = \eta_{0k} + \eta_{1k}Z_i^A + \eta_{2k}Z_i^{DD} + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i),t(i)} + \nu_{ik},$$

for key components of X_i (demographics, risk score / HCC summary, comorbidity indicators, and diagnosis/condition categories), and report standardized differences across instrument-induced variation. Under instrument validity, the excluded instruments should not systematically predict baseline risk once conditioning on location, time, plan-by-year fixed effects, and other controls. Balance is particularly relevant for the ambulance IV, where the identifying variation comes from ambulance providers with different destination propensities within the same pickup area and time period.

5.1.3 Robustness: heterogeneity by concentration bins

As a robustness check, we replace the continuous interaction in (15) with discrete concentration bins. Let $Q_m \in \{1, \dots, Q\}$ denote the market’s concentration bin (e.g., terciles) of

the predetermined M_m . We estimate

$$y_i = \beta_0 + \sum_{q=1}^Q \beta_q (D_i \times \mathbb{1}\{Q_{m(i)} = q\}) + X_i' \gamma + \alpha_{m(i)} + \tau_{t(i)} + \phi_{\text{plan}(i), t(i)} + \varepsilon_i, \quad (17)$$

instrumenting each interacted treatment $D_i \times \mathbb{1}\{Q_{m(i)} = q\}$ with the corresponding interacted instruments $Z_i^A \times \mathbb{1}\{Q_{m(i)} = q\}$ and $Z_i^{DD} \times \mathbb{1}\{Q_{m(i)} = q\}$ (and including the uninteracted instruments as needed for the level first stage). Equation (17) yields bin-specific IV effects that can be plotted against concentration bins to visualize the “fade-out” pattern without imposing linearity in M_m .

5.1.4 Plan controls and sample robustness.

Our baseline specifications include plan-by-year fixed effects to absorb plan-level utilization management and benefit design. As an additional robustness check, we re-estimate the IV models restricting the sample to beneficiaries enrolled in non-PSHP MA plans, thereby holding plan ownership constant and focusing on variation in the admitting hospital’s sponsorship status.